

Basic Biosolids Calculations

This is intended as an aid for performing basic biosolids loading calculations for record keeping and reporting purposes.
This worksheet does not relieve the permittee from the responsibility for performing accurate calculations.

1. DETERMINE DRY TONS

Enter Percent Total Solids (%TS):	3.17
Enter Specific Bulk Density (lb/cubic yard) ¹ :	0
¹ (if quantity is in cubic yards)	
Quantity	Units
Enter the biosolids quantity and units:	559763 gallons
Results:	Dry Tons: 73.994511207
Dry Metric Tons (for EPA):	67.126688451

2. DETERMINE DRY TONS PER ACRE

Enter Dry Tons:	73.994511207
Enter Acres:	71
Results:	Dry Tons Per Acre: 1.0421762142
Dry Metric Tons Per Hectare (for EPA):	2.336239156

3. DETERMINE NUTRIENT LOADINGS

Enter Dry Tons Per Acre:	1.0421762142
Enter Percent Total Nitrogen (%TN) ² :	9.89
Enter Percent Total Phosphorus (%TP) ² :	4.6
Enter Percent Total Potassium (%TK) ² :	0.47

²Use this to convert from mg/kg to % if needed

0 mg/kg = 0 %

Results:

lb/acre TN:	206.142455165
lb/acre TP:	95.8802117048
lb/acre TK:	9.7964564133

lb/acre PAN ³	137.42830344
lb/acre P ₂ O ₅	220.52448692

³ PAN based on 1.5 DEP factor. Calculations for PAN using EPA method are not provided.

kg/ha TN (for EPA):	231.054078001
kg/ha TP (for EPA):	107.467013024
kg/ha TK (for EPA):	10.9803252437

4. DETERMINE METALS LOADINGS

Enter Dry Tons Per Acre: 1.0421762142

Enter Metals Concentrations (mg/kg):

		Loading Results:	
Metal	mg/kg	Metal	lb/acre
Arsenic	29.2	As	0.0608630909
Cadmium	4.6	Cd	0.0095880212
Copper	1150	Cu	2.3970052926
Lead	68.2	Pb	0.1421528356
Mercury	13.6	Hg	0.028347193
Molybdenum	37.3	Mo	0.0777463456
Nickel	36.2	Ni	0.0754535579
Selenium	8.2	Se	0.0170916899
Zinc	3010	Zn	6.2739008094

		kg/ha (for EPA)
Arsenic	29.2	0.0682181909
Cadmium	4.6	0.0107467013
Copper	1150	2.6866753256
Lead	68.2	0.159331528
Mercury	13.6	0.031772856
Molybdenum	37.3	0.0871417301
Nickel	36.2	0.0845718668
Selenium	8.2	0.0191571632
Zinc	3010	7.0320806348